Appendix

Note: Not all formulas in this appendix are required for every problem. Select the appropriate relations based on the physical context of each question.

Physical Constants

Constant	Value
Elementary charge	$q = 1.602 \times 10^{-19}\,\mathrm{C}$
Boltzmann constant	$k = 1.381 \times 10^{-23}\,\mathrm{J/K} = 8.617 \times 10^{-5}\,\mathrm{eV/K}$
Planck constant	$h = 6.626 \times 10^{-34}\,\mathrm{J \cdot s}$
Speed of light	$c = 2.998 \times 10^{8}\,\mathrm{m/s}$
Permittivity of free space	$\epsilon_0 = 8.854 \times 10^{-14}\,\mathrm{F/cm}$
Thermal voltage at 300 K	$V_T = kT/q \approx 0.0259\,\mathrm{V}$
hc (useful for photonics)	$hc = 1.240\,\mathrm{eV} \cdot \mu\mathrm{m}$

Silicon Material Parameters at 300 K

Parameter	Value
Intrinsic carrier concentration	$n_i = 1.0 \times 10^{10}\,\mathrm{cm^{-3}}$
Relative permittivity	$\epsilon_r(\mathrm{Si}) = 11.7$
Electron mobility (typical)	$\mu_n \approx 600\text{–}1400\,\mathrm{cm^2/V \cdot s}$
Hole mobility (typical)	$\mu_p \approx 200\text{–}500\,\mathrm{cm^2/V \cdot s}$
Bandgap energy	$E_g = 1.12\,\mathrm{eV}$

Silicon Dioxide (Gate Oxide) Parameters

Parameter	Value
Relative permittivity	$\epsilon_r(\mathrm{SiO_2}) = 3.9$
Permittivity	$\epsilon_{ox} = 3.9\,\epsilon_0 \approx 3.45 \times 10^{-13}\,\mathrm{F/cm}$

Key Formulas

Carrier Statistics and Recombination

Mass action law: $n_0 p_0 = n_i^2$

Intrinsic carrier concentration: $n_i = \sqrt{N_c N_v}\exp\left(-\frac{E_g}{2k_B T}\right)$

Fermi potential (p-type): $\phi_F = \frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right)$

Fermi potential (n-type): $\phi_F = \frac{kT}{q}\ln\left(\frac{n_i}{N_d}\right)$

Excess carrier decay: $\delta n(t) = \delta n(0)\, e^{-t/\tau}$

Steady-state excess (uniform generation): $\delta n_{\text{ss}} = g\,\tau$

Minority carrier lifetime (low injection, n-type): $\tau_{p0} \approx \frac{1}{\alpha_p n_0}$

Minority carrier lifetime (low injection, p-type): $\tau_{n0} \approx \frac{1}{\alpha_n p_0}$

Typical bimolecular recombination coefficients for Si: $\alpha_n, \alpha_p \sim 10^{-15}$–$10^{-10}\,\text{cm}^3\,\text{s}^{-1}$

Drift-Diffusion and Continuity

Electron current density: $J_n = qn\mu_n\mathcal{E} + qD_n\frac{\partial n}{\partial x}$

Hole current density: $J_p = qp\mu_p\mathcal{E} - qD_p\frac{\partial p}{\partial x}$

Einstein relation: $D = \frac{kT}{q}\mu$

Ambipolar diffusion coefficient (general): $D' = \frac{D_n D_p(n_0 + p_0)}{D_n n_0 + D_p p_0}$

Ambipolar diffusion coefficient (n-type, $n_0 \gg p_0$): $D' \approx D_p$

Ambipolar diffusion coefficient (p-type, $p_0 \gg n_0$): $D' \approx D_n$

Ambipolar mobility (general): $\mu' = \frac{\mu_n\mu_p(n_0 - p_0)}{\mu_n n_0 + \mu_p p_0}$

PN Junction Diode

$$\text{Ideal diode equation:} \quad I = I_0 \left(e^{V/V_T} - 1\right)$$

$$\text{Open-circuit voltage (solar cell):} \quad V_{oc} = V_T \ln\left(1 + \frac{I_L}{I_0}\right)$$

$$\text{Depletion width (general, abrupt junction):} \quad W = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q} \cdot \frac{N_a + N_d}{N_a N_d}}$$

$$\text{Depletion width (one-sided, } N_a \gg N_d\text{):} \quad W \approx \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{qN_d}}$$

$$\text{Built-in potential:} \quad V_{bi} = \frac{kT}{q} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

Photodetectors and Optoelectronics

$$\text{Photon energy:} \quad E_{\text{photon}} = h\nu = \frac{hc}{\lambda}$$

$$\text{Photocurrent:} \quad I_{\text{ph}} = \eta\, q\, \frac{P_{\text{opt}}}{E_{\text{photon}}} = \eta\, q\, \frac{P_{\text{opt}}\lambda}{hc}$$

$$\text{Responsivity:} \quad \mathcal{R} = \frac{I_{\text{ph}}}{P_{\text{opt}}} = \eta\, \frac{q\lambda}{hc}$$

$$\text{Peak LED wavelength:} \quad \lambda_{\text{peak}} \approx \frac{hc}{E_g}$$

$$\text{Internal quantum efficiency:} \quad \eta_{\text{int}} = \frac{\tau_{nr}}{\tau_r + \tau_{nr}} = \frac{1/\tau_r}{1/\tau_r + 1/\tau_{nr}}$$

$$\text{External quantum efficiency:} \quad \eta_{\text{ext}} = \eta_{\text{int}} \times \eta_{\text{extract}}$$

Bipolar Junction Transistor (BJT)

$$\text{Current relationships:} \quad I_E = I_B + I_C, \quad I_C = \beta I_B, \quad I_C = \alpha I_E$$

$$\text{Gain conversion:} \quad \beta = \frac{\alpha}{1-\alpha}, \qquad \alpha = \frac{\beta}{\beta + 1}$$

$$\text{Shockley equation (active mode):} \quad I_C \approx I_S\, e^{V_{BE}/V_T}$$

$$\text{Early effect (base-width modulation):} \quad I_C = I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A}\right)$$

Current directions (npn, active): I_E leaves, I_B enters, I_C enters

Current directions (pnp, active): I_E enters, I_B leaves, I_C leaves

MOSFET (n-channel)

$$\text{Gate oxide capacitance per unit area:} \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$\text{Threshold voltage (simplified):} \quad V_{TH} = V_{FB} + 2\phi_F + \frac{\sqrt{4\epsilon_s q N_a \phi_F}}{C_{ox}}$$

$$\text{Body effect (substrate bias):} \quad V_{TH} = V_{TH0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$$

$$\text{Transconductance parameter:} \quad k_n = \mu_n C_{ox}\frac{W}{L}$$

$$\text{Cutoff } (V_{GS} < V_{TH}): \quad I_D = 0$$

$$\text{Linear/Triode } (V_{GS} \geq V_{TH},\, V_{DS} < V_{GS} - V_{TH}): \quad I_D = k_n\left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$$\text{Saturation } (V_{GS} \geq V_{TH},\, V_{DS} \geq V_{GS} - V_{TH}): \quad I_D = \frac{1}{2}k_n(V_{GS} - V_{TH})^2$$

$$\text{Transconductance (saturation):} \quad g_m = k_n(V_{GS} - V_{TH}) = \sqrt{2k_n I_D}$$

MOSFET (p-channel) – Source-Referenced Convention

$$\text{Cutoff } (V_{SG} < |V_{TH}|): \quad I_D = 0$$

$$\text{Linear } (V_{SG} \geq |V_{TH}|,\, V_{SD} < V_{SG} - |V_{TH}|): \quad I_D = k_p\left[(V_{SG} - |V_{TH}|)V_{SD} - \frac{V_{SD}^2}{2}\right]$$

$$\text{Saturation } (V_{SG} \geq |V_{TH}|,\, V_{SD} \geq V_{SG} - |V_{TH}|): \quad I_D = \frac{1}{2}k_p(V_{SG} - |V_{TH}|)^2$$

$$\text{where } V_{SG} = V_S - V_G, \quad V_{SD} = V_S - V_D, \quad k_p = \mu_p C_{ox} W/L$$

Unit Conversions and Useful Relations

$1\,\text{m} = 100\,\text{cm} = 10^6\,\mu\text{m} = 10^9\,\text{nm}$

$1\,\text{F/m}^2 = 10^{-4}\,\text{F/cm}^2, \qquad 1\,\text{F/cm}^2 = 10^4\,\text{F/m}^2$

$1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$

For SiO_2: $\epsilon_{ox} = 3.9 \times 8.854 \times 10^{-14}\,\text{F/cm} \approx 3.45 \times 10^{-13}\,\text{F/cm}$

Photon energy (eV) from wavelength (μm): $E(\text{eV}) = \dfrac{1.240}{\lambda(\mu\text{m})}$